

Discrete Mathematics II

List of Exercises

- Using parse trees, give a bound for the number of subformulas of a formula containing n occurrences of Boolean operators. When is this bound tight?
- Find a formula ϕ containing p, q and r such that, for every valuation $v: \{p, q, r\} \rightarrow \{1, 0\}$, changing any one of the values $v(p)$, $v(q)$ or $v(r)$ also changes $\llbracket \phi \rrbracket_v$.
- Show that the formula $\phi = \neg((q \rightarrow \neg p) \wedge (p \rightarrow (r \vee q)))$ is satisfiable but not valid.
- Give an example of a 3-element set Φ of formulas such that Φ is not satisfiable but every 2-element subset of Φ is. Generalize your example to n -element sets.
- Prove that $\Phi = \{\phi_1, \phi_2, \phi_3, \dots\}$ is satisfiable if and only if, for infinitely many n , the formula $\bigwedge_{i=1}^n \phi_i$ is satisfiable.
- Show that $\models (p \rightarrow q) \vee (q \rightarrow r)$ holds but $\models ((q \rightarrow (p \vee (q \rightarrow p))) \vee \neg(p \rightarrow q)) \rightarrow p$ does not.
- Consider the formula $\phi = (\neg p \rightarrow q) \vee ((p \wedge \neg r) \leftrightarrow q)$. Construct equivalent formulas in CNF and in DNF using the algorithm seen in class.
- Consider the formula ϕ given by the truth table

p	q	r	ϕ
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	1
1	1	0	0
1	1	1	0

Construct equivalent formulas in CNF and in DNF using the procedure seen in class.

- Consider the formulas $\phi = (\neg p_1 \vee \neg p_2 \vee \neg p_4) \wedge \neg p_5 \wedge (\neg p_3 \vee p_1) \wedge p_3 \wedge p_2 \wedge (\neg p_6 \vee p_4) \wedge p_6$ and $\psi = (\neg p_1 \vee \neg p_2 \vee \neg p_5) \wedge \neg p_5 \wedge (\neg p_3 \vee p_1) \wedge p_3 \wedge p_2 \wedge (\neg p_6 \vee p_4) \wedge p_6$, which differ only in the first clause. Use the HORNSAT algorithm seen in class to determine which of them is satisfiable (and give the model obtained) and which one is not.
- The *intersection* of two valuations v and w is defined by $(v \cap w)(p) = 1$ if and only if $v(p) = 1$ and $w(p) = 1$, for all variable p . Prove that models for Horn formulas are closed under intersection, that is, if v and w are models for a Horn formula ϕ , then also $v \cap w$ is a model for ϕ .
- Write all possible Horn clauses using only the variables p and q , in both disjunctive and implication forms, omitting tautologies and unnecessary repetitions of literals. You will obtain all disjunctions of up to two literals except for $p \vee q$, but that alone does not prove that the formula $\phi = p \vee q$ does not have an equivalent Horn formula. Instead, prove this fact using exercise 10.
- For the formula $\phi = \{\{p, \neg q, r\}, \{q, r\}, \{\neg p, r\}, \{q, \neg r\}, \{\neg r\}\}$, compute $\text{Res}(\phi)$ and $\text{Res}^2(\phi)$.
- Prove that, for every finite formula ϕ , there is a $k \geq 0$ such that $\text{Res}^k(\phi) = \text{Res}^{k+1}(\phi) = \dots = \text{Res}^*(\phi)$. Estimate k in terms of the number of clauses and the number of different variables in ϕ .
- Consider the formula $\phi = (p \vee q) \wedge (\neg p \vee r) \wedge (\neg q \vee r) \wedge (\neg r \vee s) \wedge \neg s$. Use the Davis-Putnam algorithm to show that ϕ is unsatisfiable, clearly explaining each step.

15. Consider the formula $\phi = (p \vee \neg q \vee r) \wedge (\neg p \vee q) \wedge (\neg q \vee \neg r) \wedge (p \vee r) \wedge (\neg p \vee \neg r)$. Use the DPLL algorithm to show that ϕ is satisfiable (and give a model for ϕ), clearly explaining each step.
16. Let $T(n)$ be the number of recursive calls made by DPLL on a given formula with n variables.
 - (a) Compute a bound for $T(n)$.
 - (b) Give a family of formulas for which this bound is tight.
 - (c) Compute a bound for $T(n)$ if we know unit propagation always eliminates at least one variable before branching.